5. Some chocolates were bought at Tk. 8 per piece, and sold them at Tk. 8·50 per piece and then the profit was Tk. 25. How many chocolates were bought?
6. The buying price of cloth is Tk. 125 per metre. If the shopkeeper sold it at the rate of Tk. 150 per metre, then he gained Tk. 2000 as profit. How many metres of cloth did the shopkeeper buy?
7. An item is bought at Tk. 190 and is sold at Tk. 175. What is the percentage of profit or loss ?
8. The cost price of 25 metres of cloth is equal to the selling price of 20 metres of cloth. What is the percentage of profit or loss ?
9. The buying price of 8 amloki is Tk. 5 , If 6 amloki is sold at Tk. 5, what is the percentage of profit or loss ?
10. If the selling price of a car is $\frac{4}{5}$ portion of cost price, what is the percentage of profit or loss ?
11. If an object is sold at Tk. 400, there is as much amount loss, there will be the profit amounting three times the loss, If it is sold at Tk. 480, what is the cost price of that object ?
12. If the selling price of a watch is Tk. 625, the loss is 10%. What should be the price of the watch if a profit of 10% is to be gained by selling it ?
13. Maisha bought 15 metres red ribbon at the rate of Tk. 20 per metre. The rate of VAT is Tk. 4 percent, and she gave Tk. 500 to the shopkeeper. How many taka will the shopkeeper return her ?
14. Mr. Roy is a government officer. He will go to India to visit Religious places. If Bangladeshi Tk. 1 is equal to Indian 0.63 Rupee, how much Bangladeshi taka would he need for Indian 3000 Rupee ?
15. Nilim is a service holder. His monthly basic pay is Tk. 22,250. The income tax of two lac fifty thousand taka of first slab of annual income is 0 taka. The rate of income tax of next annual income slab is Tk. 10. How many taka does he has to pay as tax ?

2·4 Problems related to speed

Speed of boat in still water and is streamy river will not be the same. In case of running boat in streamy river along with the current (down stream), the speed of stream should be added to the actual speed of boat. In case of running boat against the current up stream the speed of stream should be subtracted form the actual speed of boat. The speed with which the boat travels

along with the current or against the current is determined as the effective speed of boat.

So, effective speed of a boat along with the current (down stream) = Actual speed of the boat + Speed of stream.

Effective speed of a boat against the current (up stream) = Actual speed of the boat – Speed of the stream.

Example 20. A boat can travel 6 km per hour in still water. Against the current it needs thrice that time to travel 6 km. How long does it need to travel 50 km. when going along with the current ?

Solution : The boat can travel 6 km in 1 hour.

Against the current, the boat can travel 6 km. in (1×3) hours or 3 hours

According to the question, in 3 hours the boat travels 6 km.

$\therefore$ „ 1 „ „ „ „ $\frac{6}{3}$ „ or 2 km

Effective speed of boat against the current opposite direction = Actual speed of boat – speed of stream

$\therefore$ Speed of stream = Actual speed of boat – effective speed of boat.

= (6 – 2) km or 4 km per hour

Effective speed of the boat along with the current (same direction) = Actual speed of boat + speed of stream.

= (6 + 4) or 10 km per hour

$\therefore$ Along with the current the boat travels 10 km in 1 hour

„ „ „ „ .. 1km „ $\frac{1}{10}$ „

$\therefore$ „ „ „ „ .. 50km „ $\frac{1\times\overset{5}{\cancel{50}}}{\underset{1}{\cancel{10}}}$ „ or 5 hours

The time for travelling 50 km is 5 hours.

Example 21. There are there pipes in a cistern. The empty cistern can be filled with water by the first and the second pipe in 30 minutes and 20 minutes respectivily. The filled cistern can be completely emptied by the third pipe in 60 minutes.

(a) What part of the cistern can be filled by the third pipe in 1 second?

(b) In how minutes will the cistern be filled if the all three pipes are opened?

(c) When will the first pipe need to be closed so that the cistern be completely filled by the first and the second pipe in 18 minutes?

Solution : (a) The Third pipe in 60 minutes can vacate 1 cistern

" " " 1 " " " $\frac{1}{60}$ "

(b) The first pipe in 30 minutes can fill 1 part

" " " 1 " " " $\frac{1}{30}$ "

By the 2nd pipe in 20 minutes is filled 1 part

" " " 1 " " " $\frac{1}{20}$ "

And, By the 3rd pipe in 60 minutes is vacated 1 part

" " " 1 " " " $\frac{1}{60}$

By opening all there pipes in iminute is filled $(\frac{1}{30}+\frac{1}{20}-\frac{1}{60})$ part

$$= \frac{2+3-1}{60} \text{ part}$$

$$= \frac{4}{60} \text{ part} = \frac{1}{15} \text{ part}$$

$\frac{1}{15}$ part is filled in 1 minute

Therefore, 1 " " " $1 \times \frac{1}{15}$ "

$= 15$ inutes

Ans : 15 minutes

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(c) By the 2nd pipe in 20 minutes is filled 1 part

" " " 1 " " $\frac{1}{20}$ "

" " " 18 " " $\frac{1\times18}{20}$ "

$= \frac{9}{10}$ part

Therefore, the remaining part is $\left(1-\frac{9}{10}\right)$ part $= \frac{10-9}{10}$ part

$= \frac{1}{10}$ part

By 1st and 2nd pipe in 1 minute is filled $\left(\frac{1}{30}+\frac{1}{20}\right)$ part

$= \frac{2+3}{60}$ part $= \frac{5}{60}$

$= \frac{1}{12}$ part

To fill $\frac{1}{12}$ part time needed is 1 minute

" 1 " " " $\frac{1\times12}{1}$

" $\frac{1}{10}$ " " " $\frac{1\times12}{1\times10}$

$= \frac{6}{5}$ minuts = 1.2 minute.

Therefore, after 1.2 minutes the pipe was closed.

Example 22. The speed of 60 m. long train is 48 km per hour. How much time will the train take to cross a nearby pillar of the rail line ?

Solution : To cross the pillar the train has to pass the distance equal to its length.

48 km = 48 × 1000 m or, 48000 m.

The train crosses 48000 metre in 1 hour

" " " 1 " " $\frac{1}{48000}$ hour or $\frac{1\times60\times60}{48000}$ seconds

" " " 60 " " $\frac{1\times\cancel{60}\times\cancel{60}\times\cancel{60}}{\cancel{48000}_{\cancel{8}\,4}}$ seconds

$= \frac{9}{2}$ seconds

$= 4\frac{1}{2}$ seconds

The train will cross the pillar in $4\frac{1}{2}$ seconds.

Exercise 2·3

1. Which is the twice divided ratio of 4:9?
   (a) 2:3 (b) 4:9
   (c) 9:4 (d) 16:81
2. If A : B = 4 : 7 and B : C = 10 : 7, what is the value of C : B : A?
   (a) 49:70:40 (b) 49:40:70
   (c) 40:70:49 (d) 40:49:70

   Answer the question no. 3-4 based on following information

   30 metre cloth are distributed among Maisa, Maria and Tania in the ratio 5 : 3 : 2
3. What is the value of second quantity is successive ratio of 4:3 and 5:6?
   (a) 20 (b) 18 (c) 16 (d) 15
4. How many metre of cloth did Maisa get?
   (a) 15 (b) 9 (c) 6 (d) 3
5. How Many metre of cloth did Maria get more than Tania?
   (a) 3 (b) 4 (c) 5 (d) 6
6. What is the successive ratio of 5 : 3 and 2 : 5
   (a) 10:6:15 (b) 3:5:6 (c) 5:6:5 (d) 15:6:10
7. Which one is the fourth proporational of 3,5,15?
   (a) 20 (b) 25 (c) 30 (d) 35

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